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Final Project

Physical Activity and Its Relationship to BMI Among U.S. Adults: A NHANES Data Analysis

Research Question: Is there a significant difference in BMI across physical activity levels?

H0: "There is no significant difference in mean BMI between different physical activity levels."

HA: "There is a significant difference in mean BMI between different physical activity levels."

Chosen Dataset: <https://wwwn.cdc.gov/nchs/nhanes/>

Obesity is a major public health issue in the United States and is associated with increased risk of chronic diseases. Body Mass Index (BMI) is used to indicate the status of weight and your overall health. Physical activity is an important factor to maintaining a healthy body weight. The purpose of this study is to examine whether there is a relationship between physical activity level and BMI among U.S. adults using NHANES data. This data measures data from BMI, and physical activity to determine to overall question. The research question is: Is there a significant difference in BMI across physical activity levels? It is hypothesized that individuals with higher levels of physical activity will have lower BMI compared to those with lower physical activity levels.

This study used data from the National Health and Nutrition Examination Survey (NHANES), a nationally representative dataset collected by the CDC. The variables used in this analysis included Body Mass Index (BMI), physical activity level, age, sex, and smoking status. Data was cleaned by removing missing values and recoding categorical variables into meaningful labels. Physical activity was categorized into groups for comparison. Descriptive statistics were

calculated for key variables, and visualizations were created using R Studio. An Analysis of Variance (ANOVA) test was conducted to determine whether there were significant differences in BMI across physical activity groups.

This descriptive analysis showed variation in BMI across participants with different physical activity levels. Visualizations indicated differences in BMI distribution between groups. An ANOVA test was conducted to examine the relationship between physical activity and BMI. The results showed a statistically significant difference in BMI across physical activity groups ($F = 123, p < 0.001$). This indicates that physical activity level is significantly associated with BMI among U.S. adults.

The findings of this study suggest that physical activity is significantly associated with BMI. Individuals with different levels of physical activity tend to have different BMI values, supporting the idea that increased physical activity may contribute to healthier body weight. These results are consistent with public health research that emphasizes the importance of regular physical activity in preventing obesity. This study found a significant relationship between physical activity and BMI among U.S. adults. It proves the overall claim that increased physical activity may support a healthier BMI. Higher physical activity levels were associated with differences in BMI. These findings highlight the importance of promoting physical activity as part of public health strategies to address obesity.